

Nonlinear Models in Financial Econometrics

Problem Set — TAR, Markov Switching, Nonlinearity Tests, Parametric Bootstrap and Forecast Evaluation

FI362 — Advanced Financial Econometrics

July 10, 2026

Instructions. Six problems, one hundred marks. Show all steps. Numerical answers to four decimal places unless stated otherwise. Questions marked [★] are conceptual and should be answered in at most one paragraph. Throughout, r_t denotes the daily log return of the S&P 500 index in percent, $r_t = 100(\log P_t - \log P_{t-1})$, and $\mathcal{F}_t = \{r_t, r_{t-1}, \dots\}$.

Worked solutions begin on the page following Problem 6. Attempt every question before turning to them. A numerical answer key is given at the very end, so that you can check a result without reading the full argument.

Problem 1. The SETAR skeleton and the limit cycle (20 marks)

Consider the deterministic skeleton of a two-regime SETAR(2; 1, 1) with delay $d = 1$ and threshold $c = 1$:

$$y_t = f(y_{t-1}), \quad f(y) = \begin{cases} 0.5 + 0.6y, & y \leq 1, \\ -0.3 + 0.4y, & y > 1. \end{cases}$$

- (a) Compute the fixed point implied by each regime taken *in isolation*. Explain why neither is a fixed point of f . What are such points called?
 - (b) Show that the sufficient stationarity condition $\phi_1^{(1)} < 1$, $\phi_1^{(2)} < 1$, $\phi_1^{(1)}\phi_1^{(2)} < 1$ holds, and explain in one sentence why the process nevertheless has no stable resting point.
 - (c) Show that the skeleton admits a **period-4 limit cycle**, and compute the four cycle points exactly (as fractions) and to four decimals. Identify the regime visited at each point.
 - (d) [★] Explain what feature of S&P 500 dynamics this construction is designed to capture, and state one reason an ARMA model cannot reproduce it.
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Problem 2. Markov chain arithmetic (15 marks)

A two-state Markov switching model for r_t has $s_t \in \{1, 2\}$ ($1 = \text{calm}$, $2 = \text{turbulent}$) with transition matrix

$$P = \begin{pmatrix} 0.95 & 0.05 \\ 0.20 & 0.80 \end{pmatrix}, \quad p_{ij} = \Pr(s_t = j \mid s_{t-1} = i).$$

- (a) Compute the expected duration of each regime.
- (b) Compute the ergodic (unconditional) probabilities $\pi = (\pi_1, \pi_2)$ and verify $\pi'P = \pi'$.
- (c) Compute P^2 and interpret the entry $(2, 1)$.

- (d) [★] A colleague estimates $\hat{p}_{11} = 0.995$ and $\hat{p}_{22} = 0.60$ on 20 years of daily S&P 500 returns. Is this economically sensible? Compute the implied durations before answering.

Problem 3. One step of the Hamilton filter (20 marks)

Use P from Problem 2. Conditional on the regime, $r_t \mid s_t = j \sim N(\mu_j, \sigma_j^2)$ with

$$\mu_1 = 0.05, \quad \sigma_1 = 0.80, \quad \mu_2 = -0.10, \quad \sigma_2 = 2.50 \quad (\text{percent}).$$

At time $t-1$ the filtered probabilities are $\xi_{t-1|t-1} = (0.80, 0.20)'$. You then observe $r_t = -3.00\%$.

- (a) Compute the predicted probabilities $\xi_{t|t-1} = P' \xi_{t-1|t-1}$. Comment on the value obtained.
- (b) Compute the conditional densities $\eta_t = (f(r_t \mid s_t = 1), f(r_t \mid s_t = 2))'$.
- (c) Compute the filtered probabilities $\xi_{t|t}$ and the contribution of r_t to the log-likelihood.
- (d) Compute $\xi_{t+2|t}$.
- (e) [★] Why does a single -3% day move $\Pr(s_t = 2)$ from 0.20 to almost 1, whereas the *smoothed* probability $\Pr(s_t = 2 \mid \mathcal{F}_T)$ might be much lower?

Problem 4. The Davies problem and the parametric bootstrap (20 marks)

You test H_0 : r_t is AR(1), against H_1 : r_t is SETAR(2; 1, 1) with delay $d = 1$ and unknown threshold c .

- (a) State precisely which parameters are unidentified under H_0 , and hence why the likelihood ratio

$$\text{LR}(c) = 2[\ell_1(c) - \ell_0]$$

is not asymptotically χ^2 .

- (b) Write out, step by step, a parametric bootstrap that delivers a valid p -value for $\widehat{\text{LR}} = \sup_{c \in \mathcal{C}} \text{LR}(c)$.
- (c) Your bootstrap with $B = 1999$ replications yields $\#\{b : \text{LR}^{*(b)} \geq \widehat{\text{LR}}\} = 17$, where $\widehat{\text{LR}} = 12.40$. Report the bootstrap p -value. Compare with the (invalid) χ_2^2 decision at the 5% level.
- (d) A student, to save computer time, fixes $c = \hat{c}$ (the estimate from the real data) inside the bootstrap loop. Explain, with the direction of the bias, what goes wrong.
- (e) [★] On daily S&P 500 returns the residual bootstrap resamples $\hat{a}_t$ i.i.d. What assumption does this violate, and in which direction is the test size distorted?

Problem 5. Forecasting: mixtures, Jensen, and VaR (15 marks)

Continue with the Markov switching model of Problems 2–3. At time T the filtered probabilities are $\xi_{T|T} = (0.30, 0.70)'$.

- (a) Compute $\xi_{T+1|T}$, the point forecast $\hat{r}_{T+1|T}$, and $\text{Var}(r_{T+1} \mid \mathcal{F}_T)$, decomposing the variance into within- and across-regime parts.
 - (b) Write down the predictive density. Compute the 1% quantile of (i) a normal distribution with the same mean and variance as in part (a), and (ii) the mixture itself (the latter requires a short numerical root-find; if you cannot do it by hand, state which of the two is further left and prove it by a tail argument). Comment on the implication for VaR.
 - (c) [★] For a SETAR model, explain why $\hat{r}_{T+2|T} \neq f(\hat{r}_{T+1|T})$ and describe the correct Monte Carlo procedure in three sentences.
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Problem 6. Diebold–Mariano vs. Clark–West (10 marks)

You compare one-step-ahead forecasts of r_t from Model 1 (AR(1)) and Model 2 (SETAR(2; 1, 1)) on $P = 500$ rolling out-of-sample origins. The realised loss differentials $d_t = e_{1t}^2 - e_{2t}^2$ have $\bar{d} = -0.009$ with sample standard deviation 0.198.

- (a) Compute the Diebold–Mariano statistic and its two-sided p -value. What would you conclude?
 - (b) Explain why this conclusion is not trustworthy here.
 - (c) The Clark–West adjusted differentials $\tilde{d}_t = e_{1t}^2 - [e_{2t}^2 - (\hat{r}_{1t} - \hat{r}_{2t})^2]$ have $\bar{\tilde{d}} = 0.022$ with sample standard deviation 0.246. Compute the CW statistic and its one-sided p -value. Conclude.
 - (d) [★] In a separate exercise, the DM statistic on a *different* pair of forecasts is $\text{DM} = -2.5551$ with $P = 250$ and $h = 1$. Apply the Harvey–Leybourne–Newbold small-sample correction and report the corrected statistic.
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